

iOS PWA Layout Audit & Fixes

Reference Guide for Safari / WebKit Optimization

Context: This document contains the exact layout modifications required to fix iOS/Safari specific glitches in the Vanilla React PWA Workout Tracker. Android devices natively handle dynamic viewports, safe areas, and fixed positioning effectively, but Apple's WebKit requires explicit DOM-anchoring (`absolute` vs `fixed`) and careful management of `safe-area-inset` properties to achieve a native app feel.

Keep this document handy for when you have an iPhone available for testing.

1. Viewport Height (100dvh) Glitch & App Centering

`index.html`

The Problem: `100dvh` dynamically calculates the viewport, but in iOS PWA standalone mode, this often stutters or resolves incorrectly, leaving literal margins. Combining a `max-w-md` root with a `flex justify-center` body forces the app into a narrow, centered column instead of spanning edge-to-edge.

The Fix: Remove `100dvh` and flex centering from the body. Pin the body perfectly to the hardware bounds using `absolute inset-0`. Apply responsive `md: constraints` to the root.

CORRECTED CODE:

```
<body class="bg-lightBg dark:bg-darkBg text-black dark:text-white transition-
colors duration-300 absolute inset-0 w-full h-full overflow-hidden">

  <div id="root" class="w-full h-full flex flex-col relative bg-lightBg
dark:bg-darkBg md:max-w-md md:mx-auto md:shadow-2xl"></div>
</body>
```

2. Tab Bar Layout Jitter During Gestures

js/app.js

The Problem: Using `fixed bottom-0` anchors the tab bar to the visual viewport. When an iOS user swipes up, Safari's visual viewport compresses, causing fixed elements to jump.

The Fix: Change the tab bar to `absolute` to anchor it to the app's structural DOM (`#root`) rather than Safari's shifting viewport bounds.

CORRECTED CODE:

```
<div className="absolute bottom-0 left-0 right-0 w-full md:max-w-md mx-auto
glass border-t border-gray-200/50 dark:border-white/10 pb-safe pt-2 px-6 flex
justify-between z-[100] transition-all">
```

3. Scrollable Content Colliding with the iOS Status Bar

js/app.js

The Problem: `pt-safe` padding is applied to the scrollable content. As the user scrolls, text glides up into the unmasked safe area, overlapping with the iOS clock/battery.

The Fix: Inject a dedicated sticky masking element directly into your fragment to blur the content as it scrolls behind the notch.

CORRECTED CODE:

```
<React.Fragment>
  <div className="fixed top-0 left-0 right-0 w-full md:max-w-md mx-auto z-
[150] pt-safe bg-lightBg/95 dark:bg-darkBg/95 backdrop-blur-md"></div>
  <div className="flex-1 w-full overflow-y-auto overflow-x-hidden relative
hide-scrollbar pb-[90px]">
    <div className="w-full min-h-full pt-safe">
```

4. Double Safe-Area Padding on Floating Button

components/HomeScreen.js

The Problem: The button is positioned 90px from the bottom to clear the tab bar, but pb-safe adds an extra 34px of padding via the Home Indicator, pushing the button too high.

The Fix: Remove pb-safe since the button is already safely positioned above the navigation bar.

CORRECTED CODE:

```
<div className="fixed bottom-[90px] left-0 right-0 w-full md:max-w-md mx-auto px-5 z-40 pointer-events-none">
```

5. WorkoutScreen Escaping App Layout Bounds

components/WorkoutScreen.js & ExercisePicker.js

The Problem: fixed positions the screen relative to the viewport. On devices wider than max-w-md, the screen expands to 100vw, breaking top/bottom alignments.

The Fix: Change fixed to absolute to securely anchor it to the #root relative container.

CORRECTED CODE:

```
<div className="absolute inset-0 w-full h-full bg-lightBg dark:bg-darkBg z-[200] flex flex-col pt-safe">
```

6. Sub-Element Viewport Anchoring (Bottom Button)

components/WorkoutScreen.js

The Problem: The "Next Exercise/Finish" button attaches to the viewport via `fixed`. It will bypass the parent's width constraints and stretch across the entire screen on larger iPhones.

The Fix: Change to `absolute bottom-0`.

CORRECTED CODE:

```
<div className="absolute bottom-0 left-0 right-0 w-full bg-gradient-to-t
from-lightBg via-lightBg to-transparent dark:from-darkBg dark:via-darkBg pb-
safe pt-8 px-5 z-20">
```

7. Solid Background Masking the Dynamic Island

components/WorkoutScreen.js

The Problem: `pt-safe` on the outermost wrapper pushes the entire UI down, forcing a solid block of color over the notch and pushing the glass header below the safe area.

The Fix: Remove `pt-safe` from the wrapper. Apply it directly to the glass header with `box-content` so it expands under the notch seamlessly.

CORRECTED CODE:

```
// wrapper (remove pt-safe):
<div className="absolute inset-0 w-full h-full bg-lightBg dark:bg-darkBg z-
[200] flex flex-col">

// Header (add pt-safe and box-content):
<div className="flex-none h-14 box-content pt-safe flex items-center justify-
between px-5 glass border-b border-gray-200/50 dark:border-white/5 z-10 w-
full">
```